

$$A\underline{x} = \underline{b}$$

$A: n \times p$  Every col of  $A$  is  $n \times 1$ .

$\underline{x}: p \times 1$  There are  $p$  cols in  $A$ .

$$\underline{b}: n \times 1$$

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} x_1 - x_2 + 0x_3 + 0x_4 &= 1 \\ 0x_1 + 1x_2 + 1x_3 + 1x_4 &= 0 \\ -x_1 + 0x_2 + 1x_3 - 1x_4 &= 0 \end{aligned}$$

Rm elimination on  $\begin{pmatrix} 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ -1 & 0 & 1 & -1 & 0 \end{pmatrix}$

If last col is free: then  $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$  is a linear comb of prior cols, i.e. cols of  $A$ . There is a soln.

If last col is pivot: then  $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$  is not a linear comb of prior cols, i.e. No SOLUTION.

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} c_4 &= c_1 + c_2 \\ -c_1 - c_2 + c_4 &= 0 \\ -1c_1 - 1c_2 + 0c_3 + 1c_4 &= 0 \end{aligned}$$

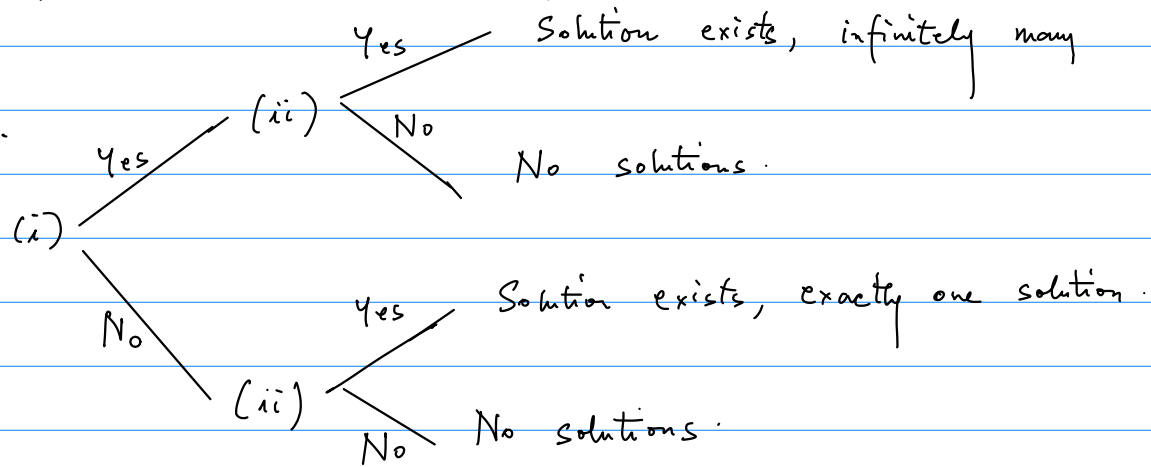
$$A \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad A\underline{x}^* = \underline{b}$$

$$A \left( \underline{x}^* + \beta \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right) = A\underline{x}^* + A\beta \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} = A\underline{x}^* + \beta \left( A \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right) = \underline{b} + \beta \underline{0} = \underline{b}$$

$$A\mathbf{x} = \mathbf{b}.$$

(i) Does  $A$  have free columns?

(ii) Is the last col a free col in  $[A \ \mathbf{b}]$ .



Suppose we found a pivot in every row of  $A$  before we exhaust all cols of  $A$ , then in  $[A \ \mathbf{b}]$ , the last col has to be free.

If  $A$  is square and we found a pivot in each row, then  $A$  has no free cols and  $A\mathbf{x} = \mathbf{b}$  has a UNIQUE soln for each  $\mathbf{b}$ .

\* If  $A$  is wide  $\left[ \begin{matrix} \# \text{rows} < \# \text{cols} \end{matrix} \right]$ , then if we find a pivot in each row, then  $A\mathbf{x} = \mathbf{b}$  has infinitely many solns no matter what  $\mathbf{b}$  is.

\* If  $A$  is tall  $\left[ \begin{matrix} \# \text{rows} > \\ \# \text{cols} \end{matrix} \right]$ , and if each col is a pivot col, then  $A\mathbf{x} = \mathbf{b}$  has  $\begin{cases} \text{unique soln if } \mathbf{b} \text{ is a linear comb of cols of } A. \\ \text{No soln if } \mathbf{b} \text{ is NOT a linear comb of cols of } A. \end{cases}$

$$A = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$$A\mathbf{x} = \mathbf{b}$$

$$A = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

$$A\underline{x} = \underline{b}$$

Since  $A$  has free cols,  $\exists$  non zero  $\underline{z}$  s.t.  $A\underline{z} = \underline{0}$

If  $A\underline{x}^* = \underline{b}$ , then  $A(\underline{x}^* + \beta \underline{z}) = \underline{b}$  for any  $\beta$ .

$$A(2\underline{x}^*) = (A2)\underline{x}^* = (2A)\underline{x}^* = 2(A\underline{x}^*) = 2\underline{b}.$$

\* Identity matrix.

$$I_{n \times n} A_{n \times m} = A_{n \times m}.$$

$$I_{3 \times 3} \underline{A}_{3 \times m} = \underline{A}_{3 \times m}.$$

$$B_{p \times n} I_{n \times n} = B_{p \times n}.$$

$$I_{3 \times 3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

\* Let  $A$  be a square matrix. ( $n \times n$ )

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 2 & 0 \end{pmatrix}.$$

Ru's Elimination

$$r_3 \leftarrow r_1 - 2r_3$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -2 \end{pmatrix} A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & -3 & 1 \end{pmatrix}$$

Swap  $r_2$  and  $r_3$ .

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -2 \end{pmatrix} A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & -3 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -2 \end{pmatrix} A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$r_1 \leftarrow 3r_1 + r_2.$$

Standard elimination.

$$r_3 \leftarrow r_3 - \frac{1}{2}r_1.$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \end{pmatrix} A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \end{pmatrix} A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 1 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \end{pmatrix} A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -2 \end{pmatrix} A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (\text{Rm's elimination}).$$

$$\begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -\frac{2}{3} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \end{pmatrix} A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (\text{Std. elimination}).$$

$$\underbrace{D U_2 U_1 S L}_B A = I. \quad \uparrow \quad (BA = I)$$

$$A = L_1^{-1} S^{-1} U_1^{-1} U_2^{-1} D^{-1}$$

$$B = D U_2 U_1 S L_1$$

$$D = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{2}{3} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad D^{-1} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{aligned} D^{-1} D &= I \\ D D^{-1} &= I \end{aligned}$$

$$U_2 = \begin{pmatrix} 1 & -\frac{2}{3} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad U_2^{-1} = \begin{pmatrix} 1 & \frac{2}{3} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{aligned} U_2^{-1} U_2 &= I \\ U_2 U_2^{-1} &= I \end{aligned}$$

$$B = D U_2 U_1 S L_1 \quad A = L_1^{-1} S^{-1} U_1^{-1} U_2^{-1} D^{-1}$$

$$BA = I$$

$$\begin{aligned} AB &= L_1^{-1} S^{-1} U_1^{-1} U_2^{-1} (D^{-1} D) U_2 U_1 S L_1 \\ &= L_1^{-1} S^{-1} U_1^{-1} U_2^{-1} U_2 U_1 S L_1 \\ &= I \end{aligned}$$

$$\begin{array}{lcl} \text{Standard :} & B & \text{Rm's} \quad C. \\ \left. \begin{array}{lcl} CAB & = & (CA)B = I \cdot B = B. \\ CAB & = & C(AB) = C. \end{array} \right\} \end{array}$$